

# Individual assignment 5

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2026-05-10

## Set up

### Packages

```
# general use and cleaning
library(tidyverse)
library(janitor)
library(readxl)
library(snakecase)

# file organization
library(here)

# arrange plots
library(patchwork)
```

### Data

```
# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))

# aquatic invertebrates
aq_ins <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Aquatic Insects")

# site information
sites <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
```

```

        sheet = "sites")

# water quality information
water_quality <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                           sheet = "Water Quality",
                           na = c("", "n/a", "N/A", "over", "173+"))

```

## Cleaning data

```

# creating a new object from sites
ncos_sites <- sites |>
  # filtering to only include NCOS quarterly sampling
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")

```

```

# creating a new object
water_quality_clean <- water_quality |>
  # cleaning column names
  clean_names() |>
  # use a filtering join, only include sites in ncos_sites
  semi_join(ncos_sites,
            by = "site") |>
  # creating new column to join with later data frames
  unite("date_site", date, site,
        remove = TRUE) |>
  # grouping by date/site
  group_by(date_site) |>
  # calculating median, pH, salinity, and DO
  summarize(med_ph = median(p_h, na.rm = TRUE),
            med_sal = median(salinity_ppt, na.rm = TRUE),
            med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
  # ungrouping data frame
  ungroup() |>
  # filtering out outliers
  filter(date_site != "2022-08-12_NVBR")

```

```

# create a new object from taxon_list
taxon_list_clean <- taxon_list |>
  # change names in verbatim_name to snake case
  mutate(verbatim_name = to_any_case(verbatim_name,
                                     case = "snake"))

```

```

# create a new object from aq_ins
aq_ins_clean <- aq_ins |>
# clean column names
clean_names() |>
# filter join to only include sites matching with ncos_sites
semi_join(ncos_sites,
          by = "site") |>
# rename column date_on_vial to date
rename(date = date_on_vial) |>
# select columns of interest
select(date, sample_type, site,
       ostracod,
       copepod,
       hemiptera_corixidae_boatman,
       cladocera,
       nematode,
       diptera_ceratopogonidae,
       annelida_oligochaete,
       diptera_chironomid,
       annelida_polychaete) |>
# filter to only include FB250/FB 250 samples
filter(sample_type %in% c("FB250", "FB 250")) |>
# pivot data frame
pivot_longer(
  # columns to make longer
  cols = ostracod:annelida_polychaete,
  # new column for the old column names
  names_to = "taxon",
  # new column for the values in each of the old columns
  values_to = "abundance"
) |>
# group by date, site, and taxon
group_by(date, site, taxon) |>
# summarize to find average abundance per liter
summarize(ave_lit = sum(abundance, na.rm = TRUE) / 7.5) |>
# ungroup
ungroup() |>
# combine date and site into date_site, don't remove date and site columns
unite("date_site", date, site, remove = FALSE) |>
# left join water_quality_clean by date_site column
left_join(water_quality_clean, by = "date_site") |>
# left join taxon_list_clean by taxon(aq_ins)/verbatim_name(taxon_list) column

```

```

left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
# mutate to create a column for full names of site codes
mutate(site_full = case_when(
  site == "NVBR" ~ "North of Venoco Bridge",
  site == "NEC" ~ "South of Venoco Bridge",
  site == "NMC" ~ "Main Channel",
  site == "NPB" ~ "South of Phelps Bridge",
  site == "NPB1" ~ "North of Phelp Bridge",
  site == "NPB2" ~ "Phelps Road",
  site == "NWP" ~ "West Pond",
  site == "NDC" ~ "Devereux Creek"
))

```

## 1. Visualizing differences across sites in major taxon abundance

### a. Plan your figure

In part 1d, you should end up with a three panel figure. All three panels should have site on the x-axis, but each panel will have a different y-axis.

In part 1c, you will create one component of the three panel figure. Provide a list of the axes and geometries you will need to create a figure in 1c showing differences between sites in ostracod abundance.

- x-axis: `site`
- y-axis: `ave_lit`
- geometry to represent average abundance/L: `geom_jitter`
- geometry to represent medians: `stat_summary`